Carbon dioxide is an atmospheric constituent that plays several vital roles in the environment. It is a greenhouse gas that traps infrared radiation heat in the atmosphere. It plays a crucial role in the weathering of rocks. It is the carbon source for plants. It is stored in biomass, organic matter in sediments and in carbonate rocks like limestone. The primary source of carbon/CO_2 is outgassing from the earth's interior at mid-ocean ridges, the decay of organic matter, hotspot volcanoes and subduction-related volcanic arcs. Photosynthesis and respiration are essentially the opposite of one another. Photosynthesis takes CO_2 from the atmosphere and replaces it with O_2. Respiration takes O_2 from the atmosphere and replaces it with CO_2.

37. In the nighttime, it is advised not to sleep under trees because :

(a) they liberate less amount of oxygen
(b) they liberate harmful gases at night
(c) they liberate carbon dioxide
(d) they liberate carbon monoxide
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (c)

Since photosynthesis does not occur at night, oxygen is not being produced by the trees. In addition to this, the trees continue respiring and releasing carbon dioxide, thereby causing the amount of carbon dioxide to be increased and the amount of oxygen to be reduced in the surrounding area. So at night if we sleep under big trees we do not have sufficient amount of oxygen.

38. It is harmful to sleep under a tree at night because the tree releases :

(a) Oxygen (b) Carbon dioxide
(c) Carbon monoxide (d) Sulphur dioxide

M.P.P.C.S. (Pre) 2000

Ans. (b)

See the explanation of above question.

39. Photosynthesis using the invisible part of the sunlight is done by some –

(a) Trees (b) Algae
(c) Bacteria (d) Fungi

U.P.P.C.S.(Pre) 2013

Ans. (c)

The bacteria which are found in the deep sea, for example green sulphur bacteria, uses infrared radiations for photosynthesis to produce energy.

II. Plant Nutrition

Notes

- Every organism is an open system linked to its environment by a continuous exchange of energy and materials.
- In ecosystems, plants and other photosynthetic autotrophs perform the crucial step of transforming inorganic compounds into organic ones.
- Plants need sunlight as the energy source for photosynthesis.
- They also need inorganic raw materials such as water, carbon dioxide (CO_2) and inorganic ions to synthesize organic molecules.
- Plants obtain CO_2 from the air. CO_2 diffuses into leaves from the surrounding air through stomata.
- Most vascular plants obtain water and minerals from the soil through their roots.
- Plants require **nine macronutrients** and at least **eight micronutrients** to sustain their life.
- Elements required by plants in relatively large quantities are called **macronutrients**.
- There are nine macronutrients in all, including the six major ingredients in organic compounds : **carbon, oxygen, hydrogen, nitrogen, sulphur** and **phosphorous**. The other three macronutrients are **potassium, calcium** and **magnesium**.
- Elements required by plants in very small amounts are called **micronutrients**.
- The eight micronutrients are **iron, chlorine, copper, zinc, manganese, molybdenum, boron** and **nickel**.
- Most of these function as cofactors, non-protein helpers in enzymatic reactions.

Macronutrients :

(i) Carbon (C)– Carbon forms the backbone of the most plant biomolecules, including proteins, starches and cellulose.

- Carbon is fixed through photosynthesis; this converts CO_2 from the air into carbohydrates, which are used to store and transport energy within the plant.

(ii) Hydrogen (H)– Hydrogen is necessary for building sugars, proteins, fats and other compounds in plant. It is obtained almost entirely from water.

- Hydrogen ions are imperative for a proton gradient to help drive the electron transport chain in photosynthesis and for respiration.

(iii) Oxygen (O)– Oxygen is a component of many organic and inorganic molecules within the plant, and is acquired in many forms as O_2 and CO_2 (mainly from the air via leaves) and H_2O, NO_3^-, $H_2PO_4^-$ & SO_4^{2-} (mainly from soil water via roots).

- Plants produce oxygen gas along with glucose during photosynthesis and require O_2 in aerobic respiration.

(iv) Nitrogen (N)– Nitrogen compounds comprise 40% to 50% of the dry matter of protoplasm. It is a constituent of amino acids (units of proteins), chlorophyll, some vitamins and nucleic acids (DNA & RNA).

- Nitrogen deficiency often results in stunted growth, slow growth, and **chlorosis**.
- The nitrogen is taken up by plants from the soil in form of NO_3^- (Nitrate ion), although in acid environments such as boreal forests, where nitrification is less likely to occur, ammonium ion NH_4^+ is the dominating source of nitrogen.

(v) Phosphorus (P)– It is the structural component of nucleic acids as well as constituents of fatty phospholipids that are important in membrane development and function. It is a part of the ATP which is the source of energy for all cells.

- Phosphorus is most commonly found in the soil in the form of polyprotic phosphoric acid (H_3PO_4) but is taken up most readily in the form of $H_2PO_4^-$.
- Plants can increase phosphorus uptake by a mutualism with mycorrhiza. A phosphorous deficiency in plants is characterized by an intense green coloration or reddening in leaves due to lack of chlorophyll.
- Phosphorus deficiency can produce symptoms similar to those of nitrogen deficiency but it differs in being extremely difficult to diagnose.

(vi) Potassium (K)– Potassium regulates the opening and closing of stomata by a potassium ion pump.

- Potassium deficiency may cause necrosis or interveinal chlorosis. It may also result in higher risk of pathogens, wilting, brown spotting and higher chances of damage from frost and heat.

(vii) Sulphur (S)– Sulphur is a structural component of some amino acids (including **cysteine** and **methionine)** and vitamins.

- It is essential for chloroplast growth and functions.
- It is found in the iron-sulphur complex of the electron transport chains in photosynthesis.
- It is needed for N_2 fixation by legumes.
- Symptoms of deficiency include yellowing of leaves and stunted growth.

(viii) Calcium (Ca)– Calcium regulates transport of other nutrients into the plant and is also involved in the activation of certain plant enzymes. It also involved in photosynthesis and plant structure.

- Calcium deficiency results in stunting. **Blossom end rot** is also a result of inadequate calcium.

(ix) Magnesium (Mg)– The outstanding role of **magnesium** in plant nutrition is as a constituent of the **chlorophyll** molecule.

- As a carrier it is also involved in numerous enzyme reactions as an effective activator, in which it is closely associated with energy-supplying phosphorus compounds.

Micronutrients (Trace elements) :

(i) Iron (Fe)– Iron is necessary for photosynthesis and is present as an enzyme cofactor in plants.

- Iron is not a structural part of chlorophyll but very much essential for its synthesis. It helps in the electron transport of plant.
- Iron deficiency can result in **interveinal chlorosis** and **necrosis**.

(ii) Molybdenum (Mo)– Molybdenum is a cofactor to enzymes important in building amino acids and is involved in nitrogen metabolism.

- It is part of the **nitrate reductase** enzyme (needed for the reduction of nitrate) and the **nitrogenase** enzyme (required for biological nitrogen fixation).

(iii) Boron (B)– Boron has many functions within a plant; it affects flowering and fruiting, pollen germination, cell division and active salt absorption.

- The metabolism of amino acids (proteins), carbohydrates, calcium, and water are strongly affected by boron.
- Boron is essential for the proper forming and strengthening of cell walls.
- Lack of boron results in short thick cells producing stunted fruiting bodies and roots. Its lack also causes failure of calcium metabolism which produces hollow heart in beets and peanuts.

(iv) Copper (Cu)– Copper is important for photosynthesis. Symptoms for copper deficiency include chlorosis.

- It is involved in many enzyme processes; involved in the manufacture of lignin (cell walls) and involved in grain production.

(v) Manganese (Mn)– Manganese is necessary for photosynthesis, including the building of chloroplast.

- Manganese deficiency may result in coloration abnormalities such as discolored spots on the leaves.

(vi) Zinc (Zn)– Zinc is required in a large number of enzymes and plays an essential role in **DNA transcription**.

- A typical symptom of zinc deficiency is the stunted growth of leaves, commonly known as 'little leaf' and is caused by the oxidative degradation of the growth hormone **auxin**.
- **Khaira** disease of rice plant (Paddy plant) is due to the deficiency of zinc.

(vii) Nickel (Ni)– In higher (vascular) plants, nickel is absorbed by plants in the form of Ni^{2+}.

- Nickel is essential for activation of **urease**, an enzyme involved with nitrogen metabolism that is required to process **urea**. Without nickel, toxic levels of urea accumulate, leading to the formation of necrotic lesions.
- In lower (non-vascular) plants, nickel activates several enzymes involved in a variety of processes.

(viii) Chlorine (Cl)– Chlorine as compounded chloride, is necessary for **osmosis** and **ionic balance**. It also plays a role in photosynthesis.

Vascular System :

- The separation between plants that have veins and plants that do not is one of the great division within the plant kingdom.
- This separates plants into **vascular** (higher) and **non-vascular** (lower) plants.
- Most plants have **xylem & phloem** and are known as **vascular** plants but some more simple plants such as algae and mosses (bryophyta), do not have xylem or phloem and are known as **non-vascular** plants.
- **Xylem :** Xylem tissue is also known as **water-conducting tissue**.
- **Xylem** is for transporting water and minerals absorbed by the roots.
- Xylem is made up of tracheids, vessels, wood parenchyma and wood fibres cells.
- **Phloem :** Phloem is responsible for transporting food from leaves (photosynthesis site) to non-photosynthesizing parts of a plant such as roots and stems.
- Phloem is also known as **bast tissue**.
- It is made up of sieve elements, companion cells, fibres, bast fibres (sclereids) and phloem parenchyma cells.

Amarbel (Cuscuta) :

- Amarbel is an **angiospermic total stem Parasite**.
- It grows over host plants with inter-twined stems, giving it a common name of **Devils' hair.**
- The plant is leafless and rootless.
- The twining stem develops **haustoria** which are rootlike and penetrate the host stem to draw water and nourishment.

Question Bank

1. Which of the following is not a correct match?

(a) Petiole : Attaches leaf to stem
(b) Thick, hard stem with branching near base : Tree
(c) Weak stem which cannot stand upright : Creeper
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

The correct matches are as follows :

Petiole	:	Attaches leaf to stem
Shrub	:	Thick, hard stem with branching near base
Tree	:	Thick, hard stem with branches high on the plant
Creeper	:	Weak stem which cannot stand upright

Hence, option (b) is the correct answer.

2. Stem of a plant helps in distributing food to all parts of the plant. It also helps in :

(a) storing the food (b) shaping the plant
(c) respiration (d) photosynthesis
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (e)

A stem provides structure to the plant. It transports food from the roots to all parts of the plants and stores the food made by the leaves. Plant stems provide both shape and support for the other parts of the plant. This is the part of the plant that bears leaves, flowers, fruits, thus also helping in their vegetative reproduction.

3. The maximum amount of water, which plants need to grow, is absorbed by the following :

(a) Embryonic zone (b) Growing point
(c) Zone of elongation (d) Root hairs

U.P.P.C.S. (Mains) 2007

Ans. (d)

The maximum amount of water is absorbed by root hairs. These are outermost layer of zone of cell maturation.

4. Which of the following statements are true?

(i) Small pores in epidermis of leaf is called stomata.
(ii) Guard cells present in the stomata are helpful in exchange of gases with atmosphere.
(iii) Transpiration takes place through stomata.

(a) (i) and (ii) (b) (i) and (iii)
(c) (ii) and (iii) (d) (i), (ii) and (iii)

Chhattisgarh P.C.S. (Pre) 2022

Ans. (d)

Stomata are the small pores present on the epidermis of leaves. The stomata consist of minute pores called stoma surrounded by a pair of guard cells. Stomata, open and close according to the turgidity of guard cells. Guard cells regulate the size of the stomatal opening. Exchange of gases with atmosphere and transpiration take place through stomata and guard cells play an important role in the working of stomata. Hence, all three statements are correct.
The closing and opening of the stomata rely on the pressure that is caused by the water's osmotic flow in the guard cells. The change in the guard cell's turgor pressure causes the stomata to open and close. Due to increased transpiration pull during the day, water is absorbed by the roots and transferred through the xylem to various areas of the plant causing the guard cell to swell and become turgid. The stomatal pore is open as a result. The guard cell shrinks and becomes stiff at night because the roots absorb less water. Consequently, stomatal pores close. So the correct option is (c).

5. The stomata open or close due to change in the :

(a) position of nucleus in cells
(b) protein composition of cells

(c) amount of water in cells
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

The closing and opening of the stomata rely on the pressure that is caused by the water's osmotic flow in the guard cells. The change in the guard cell's turgor pressure causes the stomata to open and close. Due to increased transpiration pull during the day, water is absorbed by the roots and transferred through the xylem to various areas of the plant causing the guard cell to swell and become turgid. The stomatal pore is open as a result. The guard cell shrinks and becomes stiff at night because the roots absorb less water. Consequently, stomatal pores close. So the correct option is (c).

6. Water reaches great heights in trees because of suction pull caused by :

(a) evaporation (b) absorption
(c) transpiration (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Water reaches great heights in the trees because of suction pull caused by transpiration. Transpiration is a process that involves loss of water vapour through the stomata of plants. The loss of water vapour from the plant cools the plant down when the weather is very hot, and water from the step and roots moves upwards or is 'pulled' into the leaves. On the other hand, plants absorb water and minerals from the soil via their roots but water does not move upward due to absorption. Similarly, water escapes the tree through evaporation. Hence, option (c) is the correct answer.

7. When the bark of a tree is removed in a circular fashion all around near its base, it gradually dries up or dies because –

(a) Water from soil cannot rise to aerial parts
(b) Roots are starved of energy
(c) Tree is infected by the soil microbes
(d) Roots do not receive oxygen for respiration

I.A.S. (Pre) 2011

Ans. (b)

There are two types of transportation tissues in vascular plants : (i) phloem, (ii) xylem. Phloem tissues constitute the innermost soft layer of the bark of a tree.When the bark of a tree cut in circular form near the base of its stem, flow of nutrients (photosynthetic food from green leaves) stops towards the roots because phloem which is the transporting tissue for food is damaged or absent. In this way roots of the plant gets no energy, so the plant gradually dries up or dies.

8. Water is conducted in vascular plants by-

(a) Phloem tissue (b) Parenchyma tissue
(c) Meristems (d) Xylem tissue

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The xylem transports water and soluble mineral nutrients from the roots throughout the plant. It is also used to replace water lost during transpiration and photosynthesis.

9. The 'xylem' in plants is responsible for transporting.

(a) Water (b) Oxygen
(c) Amino acid (d) Food

U.P. P.C.S. (Pre) 2023
U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

There are two types of transporting tissues found in vascular plants. (i) Phloem and (ii) Xylem. Xylem is the specialised tissue of vascular plants that transports water and minerals from plants - soil interface to stems and leaves and provides mechanical support and storage.

10. Water in plants is transported by :

(a) Xylem (b) Epidermis
(c) Phloem (d) Cambium
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

See the explanation of above question.

11. 'Phloem' in plants is mainly responsible for –

(a) Transportation of food
(b) Transportation of amino acid
(c) Transportation of oxygen
(d) Transportation of water

U.P. Lower Sub. (Pre) 2015

Ans. (a)

Phloem also called 'bast' tissues in plants that carry food produced in the leaves to all other parts of the plant. Phloem is composed of various specialized cells called sieve tubes, companion cells, phloem fibres and phloem parenchyma cells.

12. Which of the living tissues acts as the carrier of organic nutrients in higher plants ?

(a) Xylem (b) Phloem
(c) Cortex (d) Epidermis

U.P.P.C.S. (Mains) 2012

Ans. (b)

Xylem is composed of dead cells, while phloem is made up of living cells. Phloem transports food from the leaves to the rest of the plant body.

13. A nail is nailed into the trunk of a tree at a point 1 m from the soil level. After 2 years the nail will :

(a) Remain at same position (b) Move sideways
(c) Move up (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (a)

When a nail is inserted into a tree trunk, it will remain at the same relative position, even as the tree grows. Trees primarily grow in length through apical meristems located at the tips of roots and shoots, not along the trunk. The tree trunk grows wider (girth) through lateral meristems, which are located underneath the bark. The nail, once inserted, is embedded within the permanent tissues of the trunk. As the tree grows, the new growth layers are added outward, around the nail, but the nail itself remains at the same relative position even after 2 years.

14. Plants receive their nutrients mainly from :

(a) chlorophyll (b) atmosphere
(c) light (d) soil
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (d)

Plants receive their nutrients mainly from the soil. The total essential plant nutrients include seventeen different elements in which carbon, oxygen and hydrogen are absorbed from the air whereas other nutrients including nitrogen are typically obtained from the soil. Mineral nutrients are absorbed by the plants' roots when uptaking water from the soil.

15. Which of the following elements is not essential for plant growth?

(a) Sodium (b) Potassium
(c) Calcium (d) Magnesium

Uttarakhand P.C.S. (Pre) 2005

Ans. (a)

Essential elements are classified into following two categories:
(i) Macroelements (Major elements) – These are required by the plant in larger quantities e.g. Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Phosphorus (P), Potassium (K), Magnesium (Mg), Calcium (Ca) and Sulphur (S).
(ii) Microelements (Minor elements or Trace elements) – These are required by the plant in low quantities. Examples are Boron (B), Zinc (Zn), Manganese (Mn), Copper (Cu), Molybdenum (Mo), Chlorine (Cl), Iron (Fe) and Nickel (Ni).

16. Which of the following is macronutrient in relation to growth of plants?

(a) Potassium (b) Zinc
(c) Boron (d) Chlorine

Chhattisgarh P.C.S. (Pre) 2020

Ans. (a)

See the explanation of above question.

17. Which of the following elements is NOT a micronutrient in plants?

(a) Iron (b) Manganese
(c) Copper (d) Magnesium

U.P. B.E.O. (Pre) 2019

Ans. (d)

See the explanation of above question.

18. Which one of the following is not an essential micronutrient for plants?

(a) Boron (b) Zinc
(c) Sodium (d) Copper

I.A.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

19. Which of the following nutrient is most useful for enhancing oil content in mustard crop?

(a) Calcium (b) Sulphur
(c) Zinc (d) Iron

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

For mustard crop, Sulphur is the most important nutrient due to its role in metabolic processes of plant and oil synthesis in seeds.

20. Plants accept nitrogen in which form?

(a) Nitrogen gas (b) Nitrite
(c) Nitrate (d) Ammonia

U.P. R.O./A.R.O. (Mains) 2021

Ans. (c)

Nitrogen is one of the most essential nutrients for plant growth. Although nitrogen gas in the atmosphere represents the largest quantity of the nutrient in the environment, it is unavailable to most plants as a direct source. Nitrogen is normally absorbed by plants from soil in the form of nitrate (NO_3^-) ions, although in acid environments such as boreal forests, where nitrification is less likely to occur, ammonium (NH_4^+) is more likely to be the dominant source of nitrogen.

21. The plants receive Nitrogen in form of :

(a) Nitric oxide (b) Nitrate

(c) Ammonia (d) Nitride

U.P. P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

22. Nitrogen is taken by plants in the form of :

(a) Oxide (b) Nitrate

(c) Nitric acid (d) Chloride

Jharkhand P.C.S. (Pre) 2021

Ans. (b)

See the explanation of above question.

23. Which component of fertilizer is used for stimulating early growth purpose ?

(a) Potassium

(b) Nitrogen

(c) Oxygen

(d) Phosphorus

70th B.P.S.C. (Pre) 2024

Ans. (d)

For stimulating early plant growth, fertilizers containing phosphorus (P) are particularly beneficial, as they promote root development and overall vigor, while nitrogen (N) supports rapid leaf and stem growth. Phosphorus plays a crucial role in root development, which is essential for early plant establishment and nutrient uptake. It stimulates early root and plant growth, and hastens maturity. It promotes strong root systems, which are vital for anchoring plants and absorbing water and nutrients. Phosphorus is mobile within the plant, and promotes root development (Particularly the development of fibrous roots), tillering and early flowering.

24. Which of the following nutrient is necessary for 'Nodule Formation' in legumes?

(a) Nitrogen (b) Silicon

(c) Boron (d) None of the above

U.P. B.E.O. (Pre) 2019

Ans. (c)

Boron is an essential micronutrient for plants. It is essential for development and growth of new cells in plant meristem. It is necessary for 'Nodule Formation' in legumes. It is associated with translocation of sugars, starches, nitrogen and phosphorus.

25. Identify parasite in the following?

(a) Pitcher plant (b) Cuscuta

(c) Bladderwort (d) Sunflower

R.A.S./R.T.S. (Pre) 1997

Ans. (b)

Cuscuta (dodder) is a genus of about 100-170 species of yellow, orange or red parasitic plant. Dodder can be identified by its thin stems-appearing leafless, with the leaves reduced to minute scales. The dodder produces haustoria. It is used in traditional medicine as a purgative and to treat disorders of the liver, spleen and urinary tract.

26. Consider the following statements :

Assertion (A) : Cuscuta (Amarbel) is an example of parasitic angiosperm.

Reason (R) : It gets its nutrition from the leaves of the host plant.

Choose your answer from the code given below :

(a) Both (A) and (R) are true, and (R) is the correct expla nation of (A).

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(c) Assertion (A) is false, but (R) is true.

(d) Assertion (A) is true, but (R) is false.

U.P. R.O./A.R.O. (Pre) 2016

Ans. (d)

Cuscuta is a rootless, yellow coloured plant with a slender stem which twines around the stem of the host. It is an example of parasitic angiosperm. It develops haustoria which enter the host plant forming contact with xylem and phloem of the host. It absorbs the prepared food, water and minerals from the host plant.

27. Crop logging is a method of –

(a) Soil fertility evaluation

(b) Plant analysis for assessing the requirement of nutrients for crop production

(c) Assessing of crop damage

(d) Testing suitability of fertilizers

U.P.P.C.S. (Pre) 2006

Ans. (b)

Crop-logging may be of great help to determine the adequate levels of different nutrients for good plant growth and high yield. The concept of crop-logging was developed by Clements of USA for the growing of sugarcane in Hawaii, where the fertilizer requirements of this crop are greatly influenced by weather and climate. Crop-logging is a method of plant analysis for assessing the requirement of nutrients for crop production.